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**The country's crime rate moderates the relation between authoritarian predispositions and the manifestations of authoritarianism: A multilevel, multinational study**

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Running head: A multilevel model of authoritarianism

The country's crime rate moderates the relation between authoritarian predispositions and the manifestations of authoritarianism: A multilevel, multinational study

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### Abstract

We performed a multilevel, multinational test of Stenner's (2005) model on authoritarianism using the 2008 European Values Survey dataset ( $N = 55,199$ , nested in 38 nations). We focussed on the effects exerted on four authoritarian manifestations (racial intolerance, political intolerance, negative attitudes towards immigrants, and moral intolerance) by the cross-level interaction between participants' authoritarian predispositions (assessed in terms of childrearing values) and their country's crime rate. Associations between authoritarian predispositions and racial intolerance, political intolerance, negative attitudes towards immigrants, and moral intolerance were significantly stronger among participants living in countries characterised by high crime rates than those among participants living in countries with low crime rates. Limitations, implications, and future directions of this study are discussed.

Keywords: Authoritarianism, Intolerance, Threat, Moderation, Multilevel analysis,

According to the first research on the topic, authoritarianism should be regarded as the consequence of the entwinement of individual characteristics and social dynamics. Fromm (1941) argued that internal (e.g., ambivalence toward authorities) and external (e.g., cultural, social, and economic organisations of Western countries) forces interact fostering the development of an authoritarian character. According to Adorno, Frenkel-Brunswik, Levinson, and Sanford (1950), the development of an authoritarian personality depends on the mutual influence of psychodynamic factors and the present and past social and economic conditions of the nation where the individual lives. However, this approach linking personality and politics was never put into practice. Fromm's (1941) work was only theoretical, while Adorno and colleagues' (1950) research only focussed on the psychodynamics of the "authoritarian personality". The interaction between personal variables and social dynamics was often neglected, even in subsequent works, which replaced the psychodynamic perspective with other, more convincing psychological models (e.g., the social learning approach: see Altemeyer, 1996).

This neglect is apparent in studies that link threat and authoritarianism. Analyses of aggregate data have shown that authoritarian attitudes and behaviours tend to spread particularly when threat is high (Doty, Peterson, & Winter, 1991; Peterson & Gerstein, 2005; Sales, 1973). However, given that correlations identified at the aggregate level do not necessarily reflect correlations identified at the individual level, ecological data are exposed to the so-called "ecological fallacy" (Robinson, 1950). Nonetheless, some researchers working at the individual level have found the existence of a link between authoritarianism and experimentally manipulated threat. For instance, Sales and Friend (1973) manipulated threat by inducing their participants to believe that they had performed well or poorly on an anagram task presented as a measure of their ability and intelligence. Perceived failure increased participants' level of authoritarianism, whereas perceived success decreased it. More recently, Duckitt and Fischer (2003) validated a model in which experimentally manipulated societal threat (performed via a scenario in which participants' country was threatened by a societal

crisis) fostered authoritarian attitudes with the mediation of dangerous world beliefs (i.e., believing that the world is a dangerous and threatening place).

As research performed in psychology labs via the presentation of abstract scenarios may lack ecological validity, other researchers (e.g., Rickert, 1998; Stevens, Bishin, & Barr, 2006) have performed field research focussing on perceived threat instead of experimentally manipulated threat.<sup>1</sup> The most convincing model linking perceived threat to authoritarianism is the model developed and tested by Duckitt (2001; Duckitt, Wagner, du Plessis, & Birum, 2002; see Sibley & Duckitt, 2008, for a review and a meta-analysis), which predicts right-wing authoritarianism (RWA, Altemeyer, 1996). RWA, which is considered to be the standard measure of authoritarianism (Feldman, 2001), is conceived as the covariation of three attitudinal clusters: (a) authoritarian submission (i.e., a strong tendency to submit to authorities, which are perceived as established and legitimate in the society in which one lives), (b) authoritarian aggression (i.e., a general aggressiveness perceived to be positively sanctioned by established authorities and directed against various people), and (c) conventionalism (i.e., a strong tendency to adhere to social conventions, which are perceived as being endorsed by one's society and its established authorities) (Altemeyer, 1996). According to Duckitt, a complex framework of correlations among personality, social worldviews, and RWA exists. In his structural equation models, personality (assessed in terms of social conformity or Openness to experience and Conscientiousness) influenced RWA via the partial mediation of dangerous social worldviews. In turn, RWA fostered prejudice, intolerance, ethnocentrism, militarism, and related constructs.

#### “The Authoritarian Dynamic”

As stated above, in this field of studies the standard is to make reference to experimentally manipulated or perceived threat. This holds true also as concerns the model developed by Stenner (2005; Feldman & Stenner, 1997) to account for the “authoritarian dynamic,” which is one of the most fascinating approaches to authoritarianism. The cornerstone of Stenner's approach relies on the distinction between authoritarian predispositions and the manifestations of authoritarianism.

From a theoretical perspective, authoritarian predispositions are relatively stable tendencies used to solve the dilemma between the appropriate balance of group authority and uniformity and individual autonomy and diversity (Duckitt, 1989). Such predispositions reach this balance by suppressing differences and achieving uniformity. Accordingly, authoritarian predispositions, which mainly depend on individuals' education level, ethnic diversity of his/her nation, religious upbringing, and whether s/he live in a liberal democracy, should be conceived as normative worldviews about the social values of obedience versus freedom and of conformity versus differences. These predispositions direct people toward attitudes and behaviours that are aimed at fostering sameness and minimising diversity of people, beliefs, and behaviours; at glorifying, encouraging, and rewarding uniformity; and at disparaging, suppressing, and punishing differences.

Stenner (2005) measured such predispositions using responses to batteries of childrearing values, in which participants indicated the qualities they considered to be most important in raising children. According to Stenner (2005), this measure, in contrast with traditional measures of authoritarianism, is a genuine, politics-free personality measure that helps researchers to distinguish efficiently between authoritarianism and aversion both to change and to government in the economy. Moreover, this measure is satisfactory because it does not refer to specific targets, objects, events or social arrangements, and, consistent with the idea that socializing children involves basic human values and goals (Martin, 1964), because it reflects people's fundamental orientations toward authority and uniformity versus autonomy and differences.

Stenner (2005) conceived a wide array of attitudinal products of authoritarian predispositions as manifestations of authoritarianism, mainly intolerant racial, political, and moral attitudes. According to her, RWA should be considered to be a manifestation of authoritarianism and not, as often appears in the literature, as an authoritarian predisposition (see Altemeyer, 1996; Peterson & Zurbriggen, 2010). Indeed, due to its strong theoretical overlap with other manifestations of authoritarianism (i.e., Jost, Glaser, Kruglanski, & Sulloway, 2003; Whitley, 1999) the models including RWA as a predictor of prejudice and intolerance toward specific groups are somewhat



tautological: in this light “the [RWA] scale is tainted throughout by specific references to what ought to be done with minorities, dissidents, and deviants: it essentially sums the very attitudes we are endeavoring to explain” (pp. 22-23).

In short, in Stenner’s (2005) model authoritarian predispositions should be conceived as pre-existing and relatively stable worldviews about the social value of obedience and of conformity, and as the tendency to suppress difference between people, beliefs, and behaviours and to insist upon sameness. In conditions of normative threat, such authoritarian predispositions may manifest themselves in terms of authoritarian predispositions, mainly racial, political, and moral intolerance. This distinction between authoritarian predispositions and manifestations has been validated empirically. Indeed, authoritarian predispositions are stable in conditions of experimentally manipulated threat (Stenner, 2005)<sup>2</sup>, while RWA and various indices of intolerance are not (e.g., Mirisola, Roccato, Russo, Spagna, & Vieno, in press; Pettigrew, Wagner, & Christ, 2010; Sibley, Wilson, & Duckitt, 2007; Suthammanont, Petersen, Owens, & Leighley, 2010).

At the core of Stenner’s model is the idea that authoritarian predispositions influence the manifestations of authoritarianism only when there is high normative threat, i.e., when people face menaces to what makes their society one and the same. Using a number of surveys and experimental data, Stenner (2005; see also Feldman & Stenner, 1997) showed the relationship between authoritarian predispositions and several manifestations of authoritarianism (e.g., intolerance, militarism, support for the death penalty, and discrimination against various outgroups) to be strongest among participants perceiving, or experimentally facing, threat. Thus, Stenner (2005) concluded that authoritarian predispositions “may be thought of as a reasonably stable individual predisposition that *expresses itself to varying degrees* under different environmental conditions. It is activated under conditions of collective threat, especially normative threat, and yields greater ‘returns’ of racism and intolerance in response to those threat to the collective. This account allows for both an enduring individual predisposition *and* attitudes and behaviors that surge and subside under different environmental conditions” (p. 33, italics in the original).

However, even Stenner's analyses were not focused on the normative threat that characterise participants' actual environments. Thus, we still do not know what happens to the conditional authoritarian predispositions-authoritarian manifestations link when actual normative threat is taken into consideration. Even though since the 1990s multilevel modelling helped researchers to test predictive models and to consider the entwinement of individual and societal variables, multilevel studies on authoritarianism are still rare. In this paper, we performed a multilevel test of Stenner's (2005) model on authoritarianism by considering the effect exerted on four attitudinal manifestations of authoritarianism by the cross-level interaction between participants' authoritarian predispositions and a measure of an actual normative threat, their country's crime rate.

### Hypothesis

Stenner's model would pass our multilevel test if the cross-level interaction between participants' authoritarian predispositions and their country's crime rate would significantly influence their authoritarian manifestations. We took into account threat stemming from criminality both because it is the linchpin of the most recent literature on the links between threat and authoritarianism (e.g., Dallago & Roccato, 2010) and because it nicely fits into Stenner's model: indeed criminality, jeopardising the social order and the dominant social norms and values, is a powerful normative threat. Consistent with Stenner (2005), we hypothesised that if normative threat increased the manifestation of authoritarian predispositions, then authoritarian predispositions would show positive associations with the manifestations of authoritarianism among people living in countries characterised by having high crime rates and not among people living in countries characterised by low crime rates.

### Method

We performed a secondary analysis of the 2008 European Values Survey (<http://www.europeanvaluesstudy.eu/>). This dataset included 67,786 participants living in 47 countries. Aggregate data on crime rates were not available for nine of these countries; therefore, we excluded them from the analyses. We also excluded participants who did not answer the battery

of questions on childrearing attitudes (see below). The final sample included 55,199 respondents from 38 European nations.

#### *Authoritarian Predisposition Measure*

Similar to Stenner (2005), we measured participants' authoritarian predisposition using a battery on their childrearing attitudes: "Here is a list of qualities that children can be encouraged to learn at home. Which, if any, do you consider to be especially important? Please choose up to five". We took into account the following four responses: obedience and good manners (pro-trait items) and independence and imagination (con-trait items). A confirmatory factor analysis on their tetrachoric correlations matrix yielded a one-dimensional solution,  $CFI = .98$ ,  $TLI = .95$ ,  $RMSEA = .02$ , WLSMV estimator. We then recoded the choice of authoritarian options as 1 and the libertarian options as -1, and we computed a summed index of authoritarian predispositions ranging from -2 to +2.

#### *Normative Threat Measure*

We assessed normative threat by relying on country level crime data downloaded from the Eurostat (<http://epp.eurostat.ec.europa.eu>) and the WorldBank databases (<http://www.worldbank.org>). In both cases, we used the crime rates reported to the police in the year the survey was performed per 100,000 inhabitants.

#### *Authoritarian Manifestations Measures*

Based Stenner (2005; Feldman & Stenner, 1997), we computed four measures of authoritarian manifestations.

1. Racial intolerance, as the sum of the following three items: (a) "When jobs are scarce, employers should give priority to [nationality] people over immigrants"; (b) "On the list are various groups of people. Could you please mention any that you would not like to have as neighbours? People of a different race"; and (c) "On the list are various groups of people. Could you please mention any that you would not like to have as neighbours? Immigrants/foreign workers." A

confirmatory factor analysis on the items' tetrachoric correlations matrix yielded a one-dimensional solution,  $CFI = .99$ ,  $TLI = .97$ ,  $RMSEA = .05$ , WLSMV estimator.

2. We measured political intolerance by combining the answers to two values from Inglehart's (1990) battery: (a) maintaining order in the nation, and (b) protecting freedom of speech (reversed). Given that participants had to indicate their first and second choice among these options, in addition to two more items not linked with authoritarianism (fighting rising prices and giving people more say in important government decisions), we assigned a value of 1 to participants who selected the first item as their first choice, a value of 0.5 to participants who chose the first item as their second choice, and a value of 0 to participants who did not choose the item. We coded the answers to the second item the same way. Similar to Feldman and Stenner (1997), we computed our index by multiplying the two items, after reversing the second one.

3. We measured moral intolerance as the average of 3 items on the moral justifiability of homosexuality, abortion, and divorce. A confirmatory factor analysis showed good fit indexes for a one-dimensional solution,  $CFI = 1.00$ ,  $TLI = 1.00$ ,  $RMSEA = .00$ , ML estimator.

4. Since confirmatory factor analysis provided good fit indexes for a one-dimension solution,  $CFI = .99$ ,  $TLI = .98$ ,  $RMSEA = .06$ , ML estimator, we assessed attitude towards immigrants by averaging the answers to a 5-item battery: (a) Immigrants undermine [country's] cultural life; (b) Immigrants take away jobs from [country]; (c) Immigrants increase crime problems; (d) Immigrants are a strain on welfare system; and (e) Immigrants will become a threat to society. Each item had a 10-point answer scale. Higher values in the index indicated negative attitudes towards immigrants.

#### *Control variables*

We controlled for participants' socio-demographic characteristics by considering their gender, age, number of children, and marital status (we assigned a value of 1 to participants who were married or had a registered partnership). Moreover, based on Stamatel (2009), we partialled out the effects of the contextual variables showing the highest correlations with the crime rate (GDP *pro capite*, the urbanism rate, and the unemployment rate).

### Data Analysis

For each of our dependent variables, we ran four two-level hierarchical regression models using the Hierarchical Linear Modeling software (Raudenbush & Bryk, 2002). For each dependent variable, after running a preliminary unconditional model, we examined the impact exerted by participants authoritarian predispositions by partialling out the effects of our individual control variables (Model 1); moreover, we estimated the variability of this effect. In Model 2, we entered the crime rate and our contextual control variables (at the country level) to explain the variability of the effect that authoritarian predispositions had on the dependent variables.

Levels of authoritarian manifestations between individuals were modelled at level 1:

$$y_{ij} = \beta_{0j} + \beta_{1j} (\text{gender}) + \beta_{2j} (\text{age}) + \beta_{3j} (\text{number of children}) + \beta_{4j} (\text{married}) + \beta_{5j} (\text{authoritarian predispositions}) + r_{ij}$$

In this equation,  $\beta$ 's represent the impact of the individual level variables we used (gender, age, number of children, marital status, and authoritarian predispositions). The subscript  $j$  represents the countries of the participants ( $j = 1, \dots, J$ ), and the subscript  $i$  is for the individual participants ( $i = 1, \dots, N_j$ ). The random effect is represented by  $r_{ij}$ .

At level 2, the variability of the effect of authoritarian predispositions (based on Raudenbush & Bryk, 2002, we group centred them<sup>2</sup>) was modelled as a function of the crime rate both after controlling for the effects of the contextual control variables (GDP, unemployment, and urbanism) and after introducing the principal effects of those variables (expressed at the second level as the effects exerted by those variables on the variability of the intercepts: effects on  $\beta_{0j}$ ):

$$\beta_{0j} = \gamma_{00} + \gamma_{01} (\text{crime rate}) + \gamma_{02} (\text{GDP}) + \gamma_{03} (\text{unemployment}) + \gamma_{04} (\text{urbanism}) + u_{0j}$$

$$\beta_{5j} = \gamma_{50} + \gamma_{51} (\text{crime rate}) + \gamma_{52} (\text{GDP}) + \gamma_{53} (\text{unemployment}) + \gamma_{54} (\text{urbanism}) + u_{5j}$$

In these equations, the  $u$ 's represent the random coefficients. All of the errors terms of the other parameters at the individual level (gender, age, number of children, and married) in the model have been fixed.

### Results

Table 1 presents the descriptive statistics for the variables we used and their correlations.

Four preliminary unconditional models showed a significant variation of each dependent variable at the country level (see Table 2).

A model where we fixed the error terms of the slopes of our individual-level independent variables to zero (with the exception of authoritarian predispositions) showed that authoritarian predispositions positively influenced all of our dependent variables. Regarding our control variables, men were more intolerant towards immigrants and more predisposed to moral intolerance than women; moreover, our dependent variables increased with participants' age, and, with the exception of political intolerance, with the number of children. Finally, being married resulted in a positive association with moral intolerance but a negative association with intolerance toward immigrants was detected.

At the second level, regarding the intercepts, crime rates negatively affected the mean of our dependent variables (except for political intolerance). GDP was negatively correlated to racial and moral intolerance and the unemployment rate to the intolerance towards immigrants. As expected, the effects exerted by participants' authoritarian predispositions on our four authoritarian manifestations were significantly variant across individuals from different countries (see the between countries  $\beta_{5j}$  line and respective  $\chi^2$  values). The cross-level interaction between individual authoritarian predispositions and crime rates at the country level were significantly and negatively associated with all of the manifestations of authoritarianism that we examined.

However, the cross-level interaction between authoritarian predispositions and unemployment rates was only significantly associated with intolerance towards immigrants and the cross-level

interaction between authoritarian predispositions and GDP only with racial and moral intolerance. An effect size computation based on Nakagawa and Cuthill (2007) showed that crime rate resulted the strongest ecological predictor of (a) intolerance toward immigrants (crime rate's effect size = .74; unemployment rate's effect size = .53); (b) political intolerance (crime rate's effect size = .87, unemployment rate's effect size = .51); and (c) moral intolerance (crime rate's effect size = .61, urbanism rate's effect size = .41, GDP's effect size = .40). Moreover, the crime rate showed to be the only significant ecological predictor of racial intolerance, effect size = .35.

Altogether, after controlling for the other contextual control variables we used and the fixed effects on the intercepts, more than half of the variability of our dependent variables at the country level was explained by the countries crime rate.

In Table 3, the simple slopes of the effects of authoritarian predispositions on the different manifestations of authoritarianism are presented separately for the countries below the 25<sup>th</sup> percentile and above the 75<sup>th</sup> percentile of crime rates. Figure 1 shows that, as hypothesised, in countries characterised by high crime rates, the effect of authoritarian predispositions led to high authoritarian manifestations. Contrary to what was hypothesised, the relationship between authoritarian predispositions and the manifestations of authoritarianism was significant even in countries characterised by low crime rates. However, the effects were significantly stronger above the 75<sup>th</sup> percentile than below the 25<sup>th</sup> percentile.

### Discussion

According to Lewin (1936), every event is the consequence of the interaction between the state of the person and, at the same time, the state of the environment in which s/he lives. In this light, the effects of stable individual differences are moderated at least in part by the characteristics of the context (Mischel, 1968). Consistent with this idea, Mondak, Hibbing, Canache, Seligson, and Anderson (2010) recently stated that “variation in people's psychological predispositions leads them to respond differently when exposed to common environmental stimuli, and, correspondingly, that the expression of personality traits will vary by situation” (p. 90). In the field of

authoritarianism, Lavine, Lodge, Polichak, and Taber (2002) showed that this concept holds true when the characteristics of the context are manipulated in experimental designs. However, experiments are typically performed on small student samples, mainly characterised by low authoritarianism levels, and in artificial settings. Thus, such studies plausibly lead to results characterised by a controversial ecological validity (Meloan, 1993). On the other hand, studies focused on perceived threat are not completely satisfactory because high level authoritarians tend to perceive the world as being more threatening than low level authoritarians (Cohrs & Ibler, 2009; Lavine et al., 2002). To overcome this limitation, in this study we performed a multilevel test of Stenner's (2005) model on the origins of the manifestations of authoritarianism on a large and representative sample of the people living in 38 European countries.

In our analyses, individual authoritarian predispositions interacted with actual normative threat expressed in terms of the crime rate of the nation where participants lived, significantly influencing the four manifestations of authoritarianism we examined: political intolerance, moral intolerance, racial intolerance, and intolerance towards immigrants. The positive relationship between authoritarian predispositions and these manifestations of authoritarianism was significantly stronger among people living in nations with strong normative threat than among people living in nations with low normative threat. In addition, the cross-level interactions between authoritarian predispositions and other forms of actual threat did not show consistent patterns of links with our authoritarian manifestations.

At present, some studies that examine the interactive influences exerted by authoritarianism and threat on prejudice and intolerance do exist (e.g., Cohrs & Asbrock, 2008; Cohrs & Ibler, 2009); however, our research allowed us to take a relevant step further. Indeed, previous research used experimentally manipulated threat in individualistic models and not actual threat in multilevel models. Thus, the interconnection between individual and contextual variables had not been taken into account. In addition, previous research used RWA as a predictor and not as a dependent variable. Given that RWA—just like prejudice and intolerance—should be considered as a variable



that changes as a function of contextual variables (Duckitt, 2001; Mirisola et al., in press; Pettigrew, Wagner, & Christ, 2010; Sibley, Wilson, & Duckitt, 2007; Suthammanont, Petersen, Owens, & Leighley, 2010), it should be conceived as a manifestation of authoritarianism more than an authoritarian predisposition. As a consequence, these studies have been based on independent and dependent variables which were both relatively susceptible to the pressures stemming from the *hic et nunc*. On the contrary, the approach we used allowed us to predict variables lying at the attitudinal level using a cross-level interaction which involved a variable lying at the personality level. Thus, relative to traditional research into authoritarianism, it minimised the semantic overlap between *explanans* and *explanandum*, giving much more strength to the conclusions we could reach.

Our results have five main implications. The first implication concerns the nature of authoritarianism. As stated above, two main conceptions of this construct are available. Consistent with the first approaches to this topic (Adorno et al., 1950; Fromm, 1941), some researchers still regard authoritarianism as a stable personality variable (e.g. Altemeyer, 1996; Peterson & Zurbriggen, 2010). This nomothetic conception of authoritarianism has been very influential and has dominated psychological research during the past 60 years (Duncan, Peterson, & Zurbriggen, 2010). However, other researchers regard authoritarianism as an ideological variable, which may change as a function of specific life experiences (Duckitt, 2001; Duckitt & Sibley, 2010).

Recent research on RWA supports this second conception. For instance, in a five-month longitudinal study, RWA significantly increased as a function of perceived threat (Sibley, Wilson, & Duckitt, 2007), and RWA increased among low authoritarians facing a perceived (Study 1) or an experimentally manipulated (Study 2) normative threat (Mirisola et al., in press). However, we should not conclude that every nuance of authoritarianism does not belong to the personality sphere. Indeed, Stenner (2005) showed that authoritarian predispositions do not change as a consequence of perceived or experimentally manipulated threat.<sup>3</sup> Thus, we believe that our results are consistent with the idea that authoritarianism should be considered to be a stable personality variable if

conceived as an authoritarian predisposition, and as an ideological, potentially changeable, variable if manifestations of authoritarianism, including RWA, are taken into account.

The second implication of this research concerns the support our study gives to the idea that “the political effects of personality do not occur in a contextual vacuum, but instead are magnified by the presence of key precipitating or ‘activating’ features of the political environment” (Lavine et al., 2002, p. 344). Based on this idea, we should expect the explanatory power of psychological models to be highest when features of the situation experienced by participants match the content associated with the specific psychological dimension they have made salient or cognitively accessible (Funder, 1982). Accordingly, “precipitating situations make personality *relevant* and thereby strengthen the connections between predispositions and their affective, cognitive, and behavioral manifestations. In the absence of trait-situational feature matching, such predispositions are likely to remain cognitively inactive and, therefore, are relatively unlikely to influence subsequent judgments and behaviors” (Lavine et al., 2002, p. 345). Future multilevel studies aimed at testing this claim in other research topics will be interesting.

The third implication of our study concerns the type of threat that, in interaction with authoritarian predispositions, was positively associated with the manifestations of authoritarianism. The interaction between authoritarian predispositions and crime rates systematically showed positive links with the four manifestations of authoritarianism we examined, while the effects of the cross-level interactions, including the other contextual variables we used as control variables (GDP *pro capite*, unemployment rate, and urbanism rate), have not been consistent. According to the criminological literature, the GDP *pro capite* and the unemployment rate account for economic threat, while the urbanism rate accounts for threat stemming from living in a socially and physically disordered environment (Cates, Dian, & Schnepf, 2003; Walklate & Mythen, 2008). Thus, we offered strong, multilevel support for Stenner’s thesis that normative threat is the type of threat at the core of people’s authoritarian attitudes. The predictive power of other forms of menace, which are particularly efficient in predicting other constructs, such as fear of crime and the perception of

being at risk of victimisation (Russo, Roccato, & Vieno, 2011; Vieno, Roccato, & Russo, submitted), was inconsistent and substantially irrelevant.

Thus, the fourth implication of our study is that it indirectly confirmed Stenner's (2005) argument that the most efficient defence against the fear of high authoritarians is the promotion of normative social order. Hence, our results have been consistent with the literature on collective reactions to threat, according to which menaced people tend to restore a sense of control and safety by emphasising group membership and by supporting hard policies against threatening outgroups (Fritzsche, Jonas, & Keller, 2011; Huddy, Feldman, Taber, & Lahav, 2005).<sup>4</sup>

The last implication of our study concerns psychological research, even outside this field of study. According to Doise (1986), psychological phenomena can be explained on four different levels: intra-individual, inter-individual, positional, and ideological. Multilevel analyses allow researchers to take their research one step further because such analyses can be the basis of predictions that simultaneously take into account individual and contextual independent variables and their cross-level interactions. We believe that research in personality and social psychology would significantly benefit from multilevel analyses diffusion.

Our study suggested some interesting ideas for further research. First, some authors conceived authoritarianism as being a "good for the self" construct (Van Hiel & De Clercq, 2009; Dallago & Roccato, 2010; Dallago, Mirisola, & Roccato, 2011, 2012). Accordingly, authoritarianism may be considered to be an anxiety buffer (Greenberg et al., 1990), in that by definition it should promote the "oneness" and the "sameness" of the society in which people live. However, Duriez and colleagues' (2012) results recently challenged this idea, showing that authoritarianism should be considered to be a risk factor more than a protective factor against negative feelings. Somewhat consistently, Peterson and Zuckerman (2010) showed that authoritarianism may lead to negative consequences for the authoritarian him/herself. Given that we could not compare the anxiety level of our high and low authoritarian participants, our results have been compatible with both

interpretations. Future studies aimed at deepening our understanding of the “goodness for the self” of authoritarianism and at solving these inconsistencies would be fruitful.

Second, the effects exerted by our predictors on the intercepts of our manifestations of authoritarianism have been surprising, at least at a first glance. Racial intolerance, political intolerance, negative attitudes towards immigrants, and moral intolerance were higher in nations characterised by low crime rates than in nations characterised by high crime rates. The other contextual variables we entered in our multilevel model did not help to explain this effect. Consistent with Stenner (2005), our focus was on the slopes of our dependent variables, and not on their intercepts. However, it could be argued that these results have been consistent with criminological results showing that the crime rate depends, at least in part, on the degree of formal and informal social control which characterizes a specific society, and the four kinds of intolerance we focussed on can be considered as factors fostering social control (Barbagli, 2008). Future research aimed at explicitly testing this interpretation be interesting.

Third, supplementary analyses (results available upon request) showed a significant variation in the levels of authoritarian predispositions among countries,  $\chi^2(35) = 4791.87$ ,  $p < .001$ . It should be noted that this variation did not distort our results, in that none of the ecological predictors we included in this study resulted significantly connected to participants’ authoritarian predispositions. Moreover, this result did not speak counter Stenner’s (2005) conception of authoritarian predispositions as stable constructs. Indeed, according to her, authoritarian predispositions are stable within the same person, and not between geographical (and socio-political) areas. However, at present the origin of authoritarian predispositions is rather under- investigated. Future multilevel studies aimed at predicting individual and contextual variations in participants’ authoritarian predispositions could be fruitful.

Our research was based on a secondary analysis. Advantages and drawbacks of this method are well known (e.g., Kiekolt & Nathan, 1985). In our case, the use of secondary analysis made it possible to obtain inexpensive, high-quality data, gathered from a large, representative sample of

the population living in 38 European countries. This approach was particularly important because representative samples are still seldom used in psychological research. The main limitation of our study concerned the measure of our main independent variable (participants' authoritarian predispositions) on the one hand and the nature of the dependent variables we used on the other hand. As concerns the former, we had to rely on just four dichotomic items taken from an 11-item battery on childrearing values. This operationalization was consistent with that used by Stenner (2005). However, future studies using more complex measures of the main variable of Stenner's model could be fruitful. At this regard, it may be argued that authoritarian predispositions may partially overlap other well-known individual variables that have been shown to predict RWA and authoritarian manifestations (e.g., openness to experiences, need for closure, social conformity). Stenner's (2005) analyses—having shown that the effects exerted by authoritarian predispositions on authoritarian manifestations do not coincide with the effects exerted by economic, political, and social conservatism—spoke in favour of a unique role played by authoritarian predispositions. However, future studies explicitly comparing the relationship between authoritarian predispositions vs. other individual variables and authoritarian manifestations would be helpful in order to integrate this approach and other theoretical models on authoritarianism.

As concerns the latter, the four manifestations of authoritarianism we could predict were fairly consistent with those used by Stenner; however, a measure of RWA was not available in the dataset we analysed. RWA strongly correlates with the dependent variables we used (Jost, Glaser, Kruglanski, & Sulloway, 2003; Whitley, 1999); thus, we are confident with our conclusions. However, a new multilevel study aimed at predicting RWA as a function of the interaction between authoritarian predispositions and normative threat would be very interesting both *per se* and because it could help researchers explicitly link Stenner's and Altemeyer's approaches, overcoming, at the same time, the common individualistic perspective at their core.

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## Footnotes

1. The choice between experimentally manipulated and perceived threat is not crucial. Indeed, a recent meta-analysis by Onraet, Van Hienl, Dhont, and Pattyn (in press) showed that, when external threat (i.e., a kind of menace threatening both the individual and the society) is concerned, no significant differences emerge when comparing objective situational threat and subjective perceptions of threat.

2. Group mean centring removed all between-country variation in authoritarian predispositions. At the conceptual level centring at the group mean yields a pure estimate of the moderating influence that a level-2 predictor exerts on the level-1 association between two variables and can not be distorted by the presence of an interaction that involve the cluster mean of the independent variable (Enders & Tofigi, 2007). That's why Hofman and Gavin (1998) and Raudenbush (1989) recommended group mean centring when cross-level interactions are of substantive interest, as in our case.

3. Stenner's (2005) results on the stability of authoritarian predispositions do not run counter the recent results by Fischer et al. (2010), who showed terrorism salience to increase punitive and controlling authoritarian parenting practices. Indeed, from the conceptual point of view, Fischer and colleagues made reference to actual "mechanisms through which parents help their child to attain their socialization goals" (Darling & Steinberg, 1993), which are highly affected by situational cues, among which threat. Thus, the authoritarian parenting practices analysed by Fischer et al.'s (2010)—like RWA (see Duckitt & Fischer, 2003)—should be considered as manifestations of authoritarianism more than as authoritarian predispositions. On the contrary, Stenner's measure of childrearing values makes reference to stable and fundamental orientations towards authority/uniformity versus autonomy/difference, without implicating specific social, political, and contextual arrangements. From the empirical standpoint, childrearing values proved not to reflect "childrearing *practices* to which respondent were subjected, or upon which they now rely" (Stenner, 2005, p. 24, italics in the original).

4. A recent research by Hetherington and Suhay (2012) on the predictors of U.S. citizens' support on the war on terror led to results that have been inconsistent with Stenner's (2005) and ours. Indeed, these authors found that people low in authoritarian predispositions adopted more restrictive and aggressive policy stands when perceiving themselves threatened from terrorism, while the favour for repressive policies showed by people high in authoritarian predisposition did not depend on feeling vs. not feeling menaced. However, there are two differences between our and Hetherington and Suhay's research. First, the independent variable we used was not completely overlapping with their (which consisted in four forced-choice items on childrearing values such as respect for elders" versus "independence and being considerate" versus "being well-behaved). Second, the dependent variables we predicted (political, moral, racial intolerance, and intolerance toward immigrants) make reference to hostile relations between participants' ingroup and their outgroups, while those they have focussed on (e.g., endorsement of federal surveillance of phone calls and emails without court warrant, installation of video cameras in public places, and making it mandatory for citizens to carry national identity cards) make reference to problematic relations within participants' ingroup. Consequently, the inconsistency between the two studies may be explained by the differential impact that a threat may exert on ingroup- vs. outgroup oriented attitudes: Future research aimed at testing this possibility should be welcome.

Table 1.

*Between Individual and Country Level Variables: Descriptive Statistics and Correlations*

|                                   | 1      | 2     | 3     | 4     | 5     | 6     | 7     | 8     | Mean    | SD     | Min    | Max      |
|-----------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|---------|--------|--------|----------|
| <u>Individual level</u>           |        |       |       |       |       |       |       |       |         |        |        |          |
| 1. Male                           | -      |       |       |       |       |       |       |       | 0.45    | 0.50   | 0      | 1        |
| 2. Age                            | -.03** | -     |       |       |       |       |       |       | 47.28   | 17.85  | 15     | 98       |
| 3. Number of children             | -.06** | .46** | -     |       |       |       |       |       | 1.61    | 1.45   | 0      | 8        |
| 4. Married                        | .06**  | .18** |       | -     |       |       |       |       | 0.56    | 0.50   | 0      | 1        |
| 5. Authoritarian predispositions  | -.01** | .10** | .37** | .03** | -     |       |       |       | 0.35    | 1.01   | -2     | 2        |
| 6. Intolerance towards immigrants | .01    | .08** | .08** | -.01  | .12** | -     |       |       | 6.13    | 2.28   | 1      | 10       |
| 7. Political intolerance          | -.01   | .07** | .05** | .04** | .10** | .08** | -     |       | 0.40    | 0.42   | 0      | 1        |
| 8. Racial intolerance             | -.01   | .09** | .06** | .04** | .14** | .32** | .13** | -     | 1.42    | 1.03   | 0      | 4        |
| 9. Moral intolerance              | .03**  | .14** | .08** | .09** | .23** | .20** | .16** | .29** | 6.32    | 2.71   | 1      | 10       |
| <u>Country level</u>              |        |       |       |       |       |       |       |       |         |        |        |          |
| 1. Crime rate                     | -      |       |       |       |       |       |       |       | 4435.80 | 301.24 | 930.10 | 15185.54 |

## A multilevel model of authoritarianism 28

|                 |       |        |      |   |          |          |         |           |
|-----------------|-------|--------|------|---|----------|----------|---------|-----------|
| 2. GDP          | .55** | -      |      |   | 33038.34 | 25232.19 | 4076.40 | 118218.77 |
| 3. Unemployment | -.29  | -.40** | -    |   | 7.55     | 5.03     | 2.60    | 32.20     |
| 4. Urbanism     | .59** | .52**  | -.27 | - | 71.13    | 12.28    | 48.00   | 97.40     |

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Table 2.

Multilevel Correlates of Authoritarian Manifestations

|                                            | Intolerance towards immigrants |       |                 |       |                 |       | Political Intolerance |       |                  |       |                |       | Racial Intolerance  |       |                 |       |                 |       | Moral Intolerance   |       |                 |       |                 |       |
|--------------------------------------------|--------------------------------|-------|-----------------|-------|-----------------|-------|-----------------------|-------|------------------|-------|----------------|-------|---------------------|-------|-----------------|-------|-----------------|-------|---------------------|-------|-----------------|-------|-----------------|-------|
|                                            | Unconditional Model            |       | Model 1         |       | Model 2         |       | Unconditional Model   |       | Model 1          |       | Model 2        |       | Unconditional Model |       | Model 1         |       | Model 2         |       | Unconditional model |       | Model 1         |       | Model 2         |       |
|                                            | Coef.<br>(SE)                  | t     | Coef.<br>(SE)   | t     | Coef.<br>(SE)   | t     | Coef.<br>(SE)         | t     | Coef.<br>(SE)    | t     | Coef.<br>(SE)  | t     | Coef.<br>(SE)       | t     | Coef.<br>(SE)   | t     | Coef.<br>(SE)   | t     | Coef.<br>(SE)       | t     | Coef.<br>(SE)   | t     | Coef.<br>(SE)   | t     |
| Intercept                                  | 6.08**<br>(.12)                | 50.62 | 5.65**<br>(.12) | 46.94 | 5.65**<br>(.11) | 53.69 | .40**<br>(.02)        | 22.81 | .29**<br>(.02)   | 14.72 | .30**<br>(.02) | 15.44 | 1.39**<br>(.05)     | 21.27 | 1.09**<br>(.05) | 15.53 | 1.09**<br>(.05) | 22.15 | 6.13**<br>(.23)     | 26.86 | 4.64**<br>(.28) | 16.53 | 4.63**<br>(.17) | 26.85 |
| <u>Individual Level</u>                    |                                |       |                 |       |                 |       |                       |       |                  |       |                |       |                     |       |                 |       |                 |       |                     |       |                 |       |                 |       |
| Male                                       |                                |       | .09**<br>(.03)  | 3.27  | .09**<br>(.03)  | 3.27  |                       |       | -.01<br>(.01)    | -1.29 | -.01<br>(.01)  | -1.32 |                     |       | .02<br>(.01)    | 1.64  | .02<br>(.01)    | 1.64  |                     |       | .26**<br>(.04)  | 6.68  | .26**<br>(.04)  | 6.66  |
| Age                                        |                                |       | .01**<br>(.01)  | 6.24  | .01**<br>(.01)  | 6.23  |                       |       | .01**<br>(.01)** | 9.88  | .01**<br>(.01) | 9.89  |                     |       | .01**<br>(.01)  | 9.72  | .01**<br>(.01)  | 9.72  |                     |       | .02**<br>(.01)  | 13.11 | .02**<br>(.04)  | 13.11 |
| Number of children                         |                                |       | .03**<br>(.01)  | 2.85  | .03**<br>(.01)  | 2.86  |                       |       | .01<br>(.01)     | 1.30  | .01<br>(.01)   | 1.32  |                     |       | .02**<br>(.01)  | 2.61  | .02**<br>(.01)  | 2.61  |                     |       | .12**<br>(.01)  | 7.66  | .11**<br>(.02)  | 7.68  |
| Married                                    |                                |       | -.10**<br>(.03) | -2.91 | -.10**<br>(.03) | -2.91 |                       |       | .02*<br>(.01)    | 2.01  | .01<br>(.01)   | 1.61  |                     |       | -.01<br>(.01)   | -1.24 | -.01<br>(.01)   | -1.24 |                     |       | .12**<br>(.03)  | 3.79  | .12**<br>(.03)  | 3.78  |
| Authoritarian Predispositions              |                                |       | .25**<br>(.04)  | 6.22  | .25**<br>(.03)  | 9.17  |                       |       | .04**<br>(.01)   | 6.79  | .04**<br>(.01) | 10.85 |                     |       | .09**<br>(.01)  | 6.26  | .09**<br>(.02)  | 8.54  |                     |       | .40**<br>(.04)  | 10.68 | .40**<br>(.03)  | 14.55 |
| <u>Country Level</u><br>(β <sub>0j</sub> ) |                                |       |                 |       |                 |       |                       |       |                  |       |                |       |                     |       |                 |       |                 |       |                     |       |                 |       |                 |       |
| Crime rate                                 |                                |       |                 |       | -.01*<br>(.01)  | -2.24 |                       |       |                  |       | -.01<br>(.01)  | -1.57 |                     |       |                 |       | -.01**<br>(.01) | -3.72 |                     |       |                 |       | -.01**<br>(.01) | -5.97 |
| GDP                                        |                                |       |                 |       | -.01<br>(.01)   | -1.72 |                       |       |                  |       | -.01<br>(.01)  | -.78  |                     |       |                 |       | -.01**<br>(.01) | -3.68 |                     |       |                 |       | -.01**<br>(.01) | -2.83 |
| Unemployment                               |                                |       |                 |       | -.05**<br>(.02) | -3.14 |                       |       |                  |       | .01<br>(.01)   | .37   |                     |       |                 |       | -.01<br>(.01)   | .64   |                     |       |                 |       | -.01<br>(.01)   | -.02  |
| Urbanism                                   |                                |       |                 |       | .02<br>(.01)    | 1.79  |                       |       |                  |       | .01<br>(.01)   | .70   |                     |       |                 |       | .01<br>(.01)    | .24   |                     |       |                 |       | .01<br>(.01)    | .33   |
| <u>Country Level</u><br>(β <sub>1j</sub> ) |                                |       |                 |       |                 |       |                       |       |                  |       |                |       |                     |       |                 |       |                 |       |                     |       |                 |       |                 |       |
| Crime rate.                                |                                |       |                 |       | .01**<br>(.01)  | 4.39  |                       |       |                  |       | .01**<br>(.01) | 5.26  |                     |       |                 |       | .01*<br>(.01)   | 2.03  |                     |       |                 |       | .01**<br>(.01)  | 3.60  |
| GDP                                        |                                |       |                 |       | .01<br>(.01)    | 1.45  |                       |       |                  |       | .01<br>(.01)   | 1.02  |                     |       |                 |       | .01<br>(.01)    | 1.39  |                     |       |                 |       | .01*<br>(.01)   | 2.15  |
| Unemployment                               |                                |       |                 |       | -.01*<br>(.02)  | -2.41 |                       |       |                  |       | -.01*<br>(.02) | -2.35 |                     |       |                 |       | -.01<br>(.01)   | -1.01 |                     |       |                 |       | .01<br>(.01)    | .13   |
| Urbanism                                   |                                |       |                 |       | -.01<br>(.01)   | -.90  |                       |       |                  |       | -.01<br>(.01)  | -1.53 |                     |       |                 |       | .01<br>(.01)    | .85   |                     |       |                 |       | -.01*<br>(.01)  | -2.39 |
| <u>Var. Comp.</u>                          |                                |       |                 |       |                 |       |                       |       |                  |       |                |       |                     |       |                 |       |                 |       |                     |       |                 |       |                 |       |
| Within Country                             | 4.682                          |       | 4.509           |       | 4.509           |       | 0.160                 |       | 0.156            |       | 0.156          |       | 0.882               |       | 0.848           |       | 0.848           |       | 5.468               |       | 4.906           |       | 4.906           |       |
| Between                                    | 0.531                          |       | 0.521           |       | 0.440           |       | 0.011                 |       | 0.011            |       | 0.009          |       | 0.158               |       | 0.159           |       | 0.081           |       | 1.926               |       | 1.998           |       | 0.609           |       |



## A multilevel model of authoritarianism 30

|                                 |           |           |           |           |           |           |           |            |           |            |            |           |
|---------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|------------|-----------|------------|------------|-----------|
| Country $\beta_{0j}$            |           |           |           |           |           |           |           |            |           |            |            |           |
| $\chi^2$                        | 5608.75** | 5780.08** | 4170.83** | 3745.24** | 3796.35** | 2614.36** | 9908.10** | 10046.20** | 4502.02** | 17388.77** | 19449.39** | 5212.92** |
| Between<br>country $\beta_{sj}$ |           | 0.059     | 0.029     |           | .0009     | .0003     |           | 0.008      | 0.004     |            | 0.050      | 0.028     |
| $\chi^2$                        |           | 615.37**  | 292.17**  |           | 271.60**  | 112.32**  |           | 452.16**   | 260.21**  |            | 498.48**   | 265.98**  |
| <i>Explained<br/>variance</i>   |           |           |           |           |           |           |           |            |           |            |            |           |
| Individual                      |           | 3.7%      |           |           | 2.5%      |           |           | 3.4%       |           |            | 10.3%      |           |
| Country $\beta_{0j}$            |           |           | 17.1%     |           |           | 18.2%     |           |            | 49.1%     |            |            | 68.4%     |
| Country $\beta_{sj}$            |           |           | 50.8%     |           |           | 66.6%     |           |            | 50.0%     |            |            | 44.0%     |

Table 3.

*Single Slopes Analysis*

|                                | Crime rate        |       | Crime rate        |       |
|--------------------------------|-------------------|-------|-------------------|-------|
|                                | < 25th percentile |       | > 75th percentile |       |
|                                | $\beta$           | $p$   | $\beta$           | $p$   |
|                                | (95% <i>CI</i> )  |       | (95% <i>CI</i> )  |       |
| Intolerance towards immigrants | .07               | <.001 | .46               | <.001 |
|                                | (.18-.22)         |       | (.42-.49)         |       |
| Political intolerance          | .02               | <.001 | .05               | <.001 |
|                                | (.01-.03)         |       | (.04-.06)         |       |
| Racial intolerance             | .06               | <.001 | .18               | <.001 |
|                                | (.05-.08)         |       | (.16-.19)         |       |
| Moral intolerance              | .44               | <.001 | .74               | <.001 |
|                                | (.40-.50)         |       | (.70-.78)         |       |

Figure caption.

*Figure 1.* Cross level interaction between authoritarian predispositions (individual level) and crime rate (country level)

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Figure 1.

